HowToDIV
Multimodal dialogue + egocentric video

Google Research

From Videos to Conversations: Egocentric Instructions for Task Assistance

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What's the Problem Statement ?

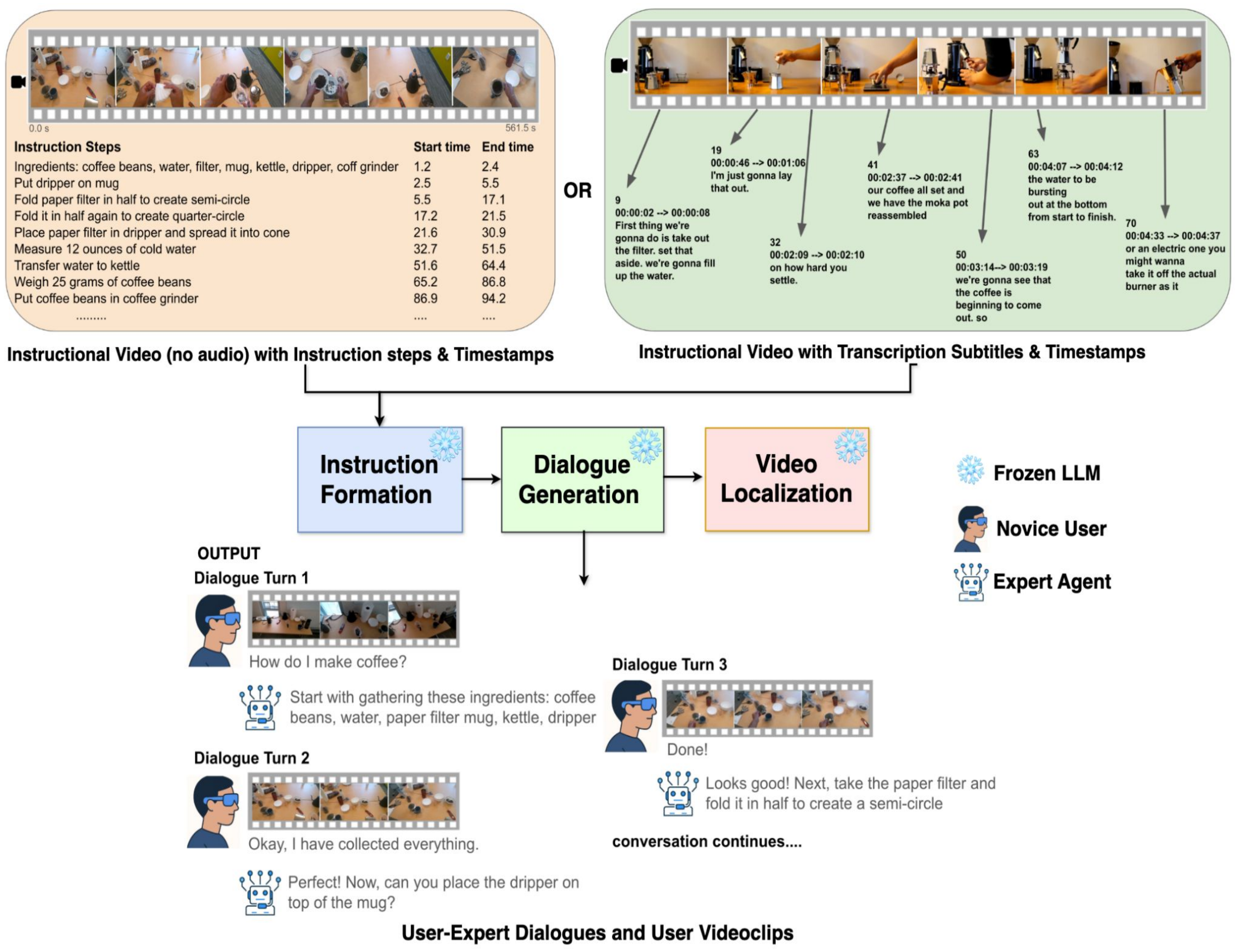
Developing interactive AR agents needs datasets but multimodal conversations don't exist at scale

- In the world of AR wearables, there is an increasing demand for task-assistant agents that guide users step-by-step while tracking progress, correcting errors, driven from an egocentric POV.
- Training such agents requires multimodal conversational interactions which are expensive to collect at scale.
- But instructional videos are abundant! Thus we present a one-shot tunable pipeline that converts single-speaker instructional videos into life-like conversations containing both transcript & video clips.

Instructional monologue video → **Grounded expert-novice dialogue for multimodal task assistance**

HowToDIV: A Dataset of Dialogues, Instructions, Videos for How-To tasks

Converting Monologue tutorials into Conversation: A fully automatic, prompt-based pipeline that converts instructional videos into multi-turn conversations with turn-level video grounding.

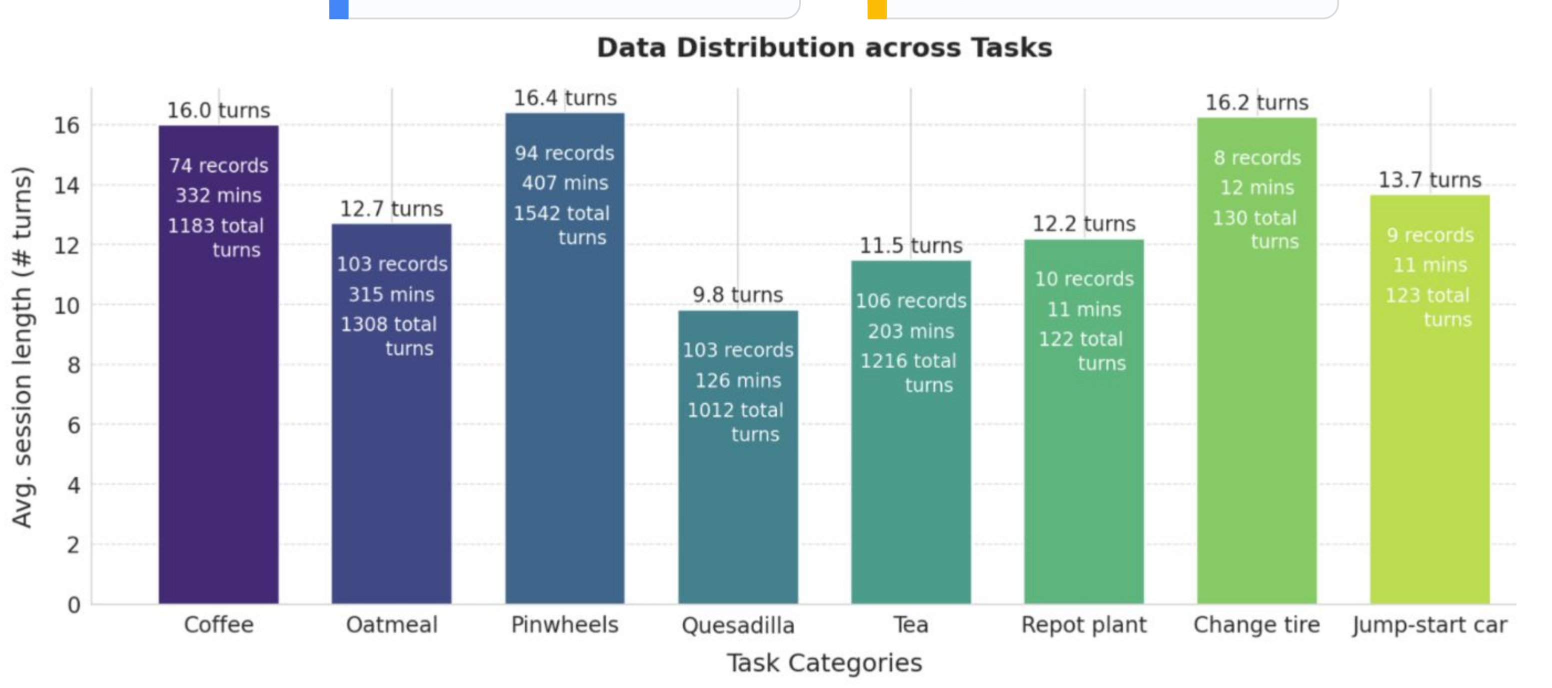


1. Instruction Formation: Recover an ordered sequence of atomic, independently executable task steps from subtitles or action annotations.
2. Dialogue Generation: Map and transform each step to a novice query and expert response (with clarifications, errors, and corrections).
3. Video Localization: Align each user turn with the corresponding egocentric video segment using timestamps or subtitle boundaries.

Using this pipeline we develop HowToDIV dataset!

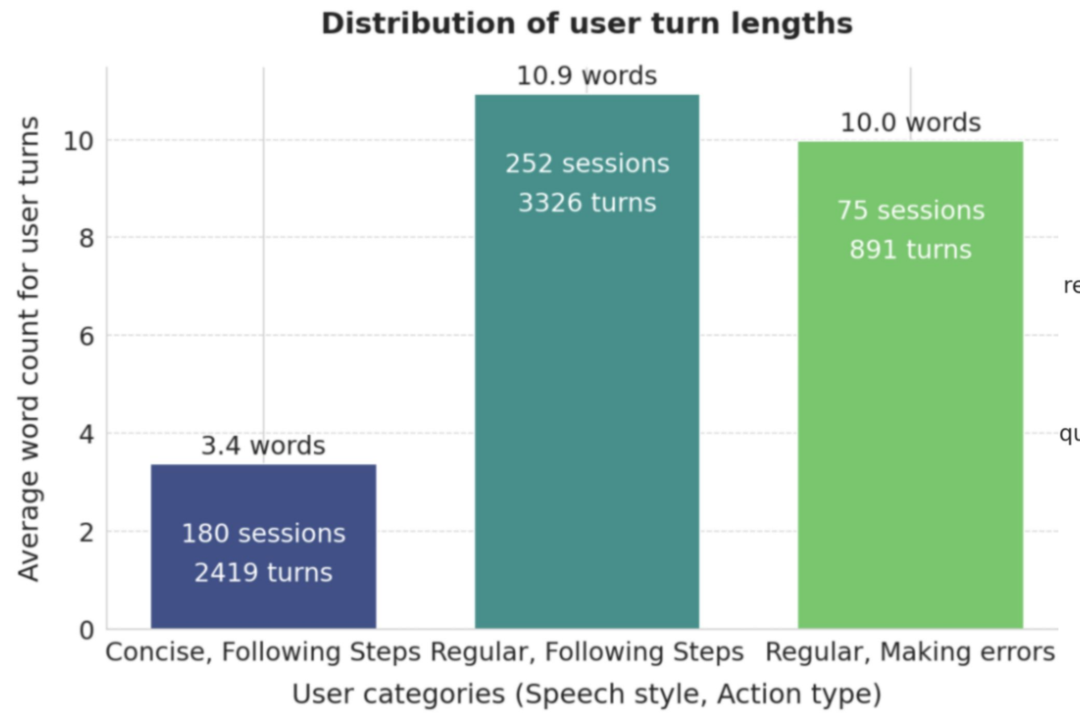
HowToDIV Dataset at a glance

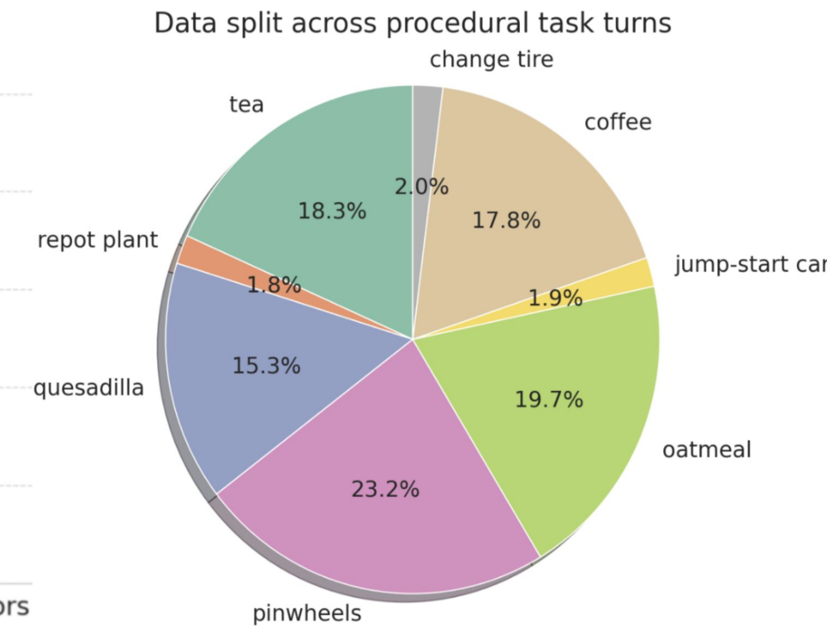
507 conversations	**6,636** question–answer pairs	**24 h** egocentric video
3 domains	**9** procedural tasks	



[1] Amazon, [2] Work done while at Google, [3] Meta

Realistic conversational variation





What makes HowToDIV unique ?

- Atomic Procedural steps - Each action is independently executable and queryable.
- Turn-level egocentric grounding - Every turn is paired with temporally aligned clip.
- Multi-turn dialogues - Conversations model progress, questions, guidance.
- Realistic human errors and corrective instructions from Expert agent.

Quality control: 93.2% of turns judged usable <4% of turns removed by automatic filters

INSTRUCTION	USER ERROR	INSTRUCTION	MODIFICATION
Place the tea bag in the mug	Dropped the tea bag on floor	Drizzle honey in bowl	Pour sugar instead of honey
Roll the tortilla into a wrap	Folded the tortilla in half	Stir using spoon	Stir using knife
Fold the filter to create a semicircle	Tore the paper filter	Slice tortilla using knife	Rip tortilla by hands
		Place tortilla on cutting board	Place tortilla on table

Table 1. Examples of (Left) User errors and (Right) Modifications and the corresponding instruction steps from HowToDIV.

Comparison with existing Datasets

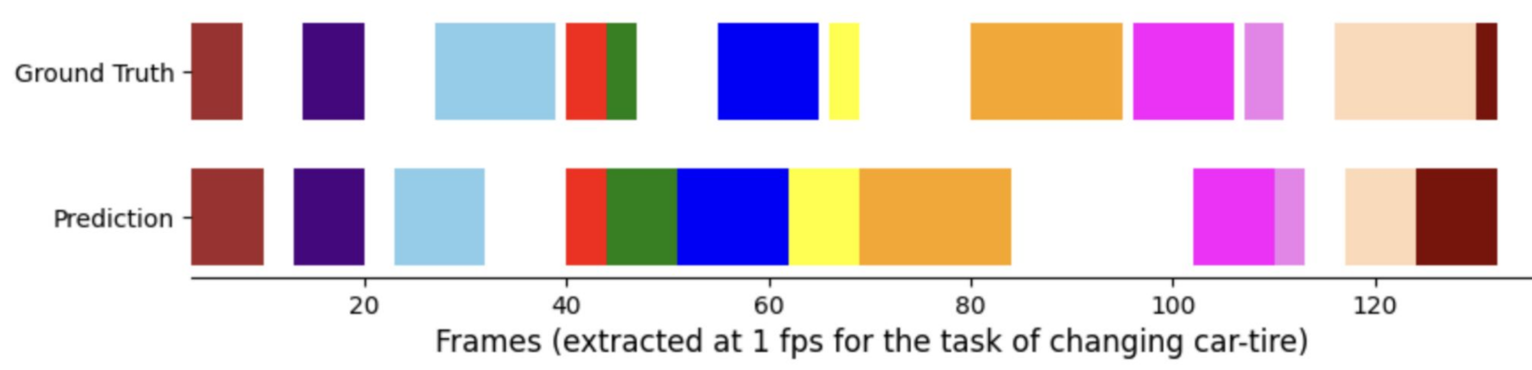
Dataset	Domain	Dialogues	Temporal-steps	Narrations	Instructions	Errors	Egocentric	Videos
NIV [10]	Varied	✗	✗	✓	✗	✗	✗	6.66
HowTo100M [20]	Varied	✗	✗	✓	✗	✗	✗	134.4k
HoloAssist [3]	Assembly	✓	✓	✗	✓	✓	✓	2221
EpicTent [4]	Tent Making	✗	✓	✗	✓	✓	✓	7
EgoPER [11]	Cooking	✗	✓	✗	✓	✓	✓	28
EGTEA [5]	Cooking	✗	✓	✗	✗	✗	✓	28
Assembly101 [16]	Toy-assembly	✗	✓	✗	✓	✓	✓	167
COIN [19]	Varied	✗	✓	✓	✗	✗	✗	476
CrossTask [17]	Varied	✗	✓	✓	✓	✗	✗	376
YouCook2 [13]	Cooking	✗	✓	✓	✓	✗	✗	176
EPIC-KITCHENS [18]	Cooking	✗	✓	✓	✗	✗	✓	100
HowToDIV (Ours)	Varied	✓	✓	✓	✓	✓	✓	576

Explicit structure improves results

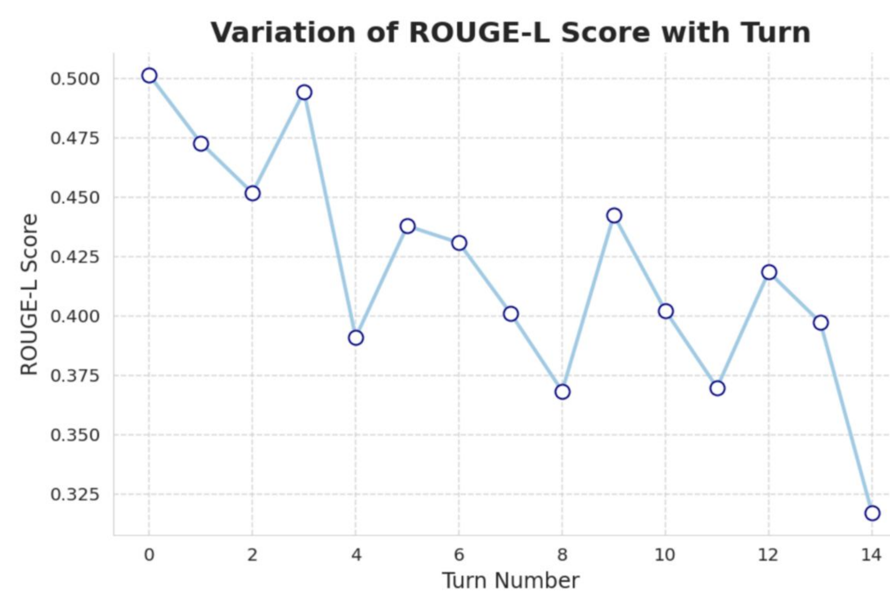
We benchmark two popular VLMs on the HowToDIV benchmark :

Method	BLEU	LLM-Judge	ROUGE-2	ROUGE-L
Gemma-3-4B (History-only)	0.321	2.870	0.125	0.270
Gemma-3-4B (History + Steps)	**0.457**	4.101	**0.268**	**0.429**
Qwen2.5-VL-7B (History-only)	0.281	2.995	0.105	0.220
Qwen2.5-VL-7B (History + Steps)	0.441	**4.232**	0.199	0.356

Key takeaway: Providing explicit procedural steps raises LLM-Judge from 2.87 → 4.10 (Gemma) and 3.00 → 4.23 (Qwen).



Temporal action localization for changing car tire; each color denotes a step



Long-horizon procedural reasoning remains challenging as errors compound in later turns

Takeaway

HowToDIV offers a new benchmark for real-time task assistance and a scalable path from abundant instructional videos to grounded conversations.

Paper Code